

# Deep-hole rigidity of the Clebsch hexagon code

Tavis Rudd

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## Abstract

Among the six-arcs of  $\text{PG}(2, 11)$ , exactly one projective class has the maximum-distance syndrome directions of its associated MDS code contained in a conic: the Clebsch hexagon. This coding-theoretic condition recovers its projective stabilizer  $A_5$ . More generally, containment of the syndrome locus in the zero set of any nonzero form of degree at most three characterizes the same class. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the core conic-rigidity implication. Enumeration proves the low-degree strengthening, the sharp numerical gaps, and the terminal  $k = 7$  exclusions.

The arc lies on no conic, so its  $[6, 3, 4]_{11}$  MDS code is non-GRS; the code has covering radius three. Its twelve projective deep-hole directions are the  $\mathbb{F}_{11}$ -points of a conic. Among nonzero syndromes outside the six arc columns, the quadratic test separates exactly the deep holes; together with the zero-syndrome and arc-column cases, it gives the complete distance oracle of Proposition 3.5. Its ambiguity data recover the classical Brianchon points, self-polar triangles, and an invariant support bipartition. The chord identity and the Sylvester graph show that  $q = 11$  is the only conic-filling field for six-arcs. A universal chord-defect identity shows that conic filling by a  $k$ -arc forces  $q < \binom{k}{2}$ . Through seven points the only cases are a four-frame in  $\text{PG}(2, 5)$  and the Clebsch hexagon; for eight points only  $q \in \{13, 16, 17, 19\}$  remain.

**Keywords.** Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic.

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## 1 Introduction

Let  $C$  be a linear  $[n, k, d]_q$  code with covering radius  $\rho$  and full-rank parity-check matrix  $H$ . For a received word  $v$ , write  $\sigma(v) = Hv^\top$  and

$$w(s) = \min\{\text{wt}(e) : He^\top = s\}.$$

Then  $d(v, C) = w(\sigma(v))$ , and  $v$  is a *deep hole* precisely when this minimum is  $\rho$ . The syndrome  $s$  specifies one coset of  $C$ , whose coset leaders are its weight- $w(s)$  representatives. Because  $w(\lambda s) = w(s)$  for  $\lambda \neq 0$ , maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\text{proj}}(C) = \{[s] \in \text{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents  $q - 1$  distinct nonzero syndrome cosets, while each coset contains  $|C|$  received words.

For an MDS code of redundancy  $r = n - k$ , the parity-check columns form a projective arc in  $\text{PG}(r - 1, q)$ , and  $w(s)$  is the least number of columns whose span contains  $[s]$ . Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by  $r - 1$  existing columns. In the redundancy-three plane case

used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write  $\mathcal{U}(A)$  for this extension locus. If  $\mathcal{U}(A) \neq \emptyset$ , then the associated code has covering radius three and  $\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C)$ . This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [7]; their Theorem 6.3 supplies the leader count used in Section 3. See also Hirschfeld [14] for the plane arc/covering-radius setting.

The paper studies one small exception to the GRS picture: a non-GRS  $[6, 3, 4]_{11}$  MDS code whose projective deep-hole locus is exactly the  $\mathbb{F}_{11}$ -point set of a nonsingular conic. Its parity-check columns form the *Clebsch hexagon*, a six-arc in  $\text{PG}(2, 11)$ . Thus a natural property of the covering-radius structure, with no assumed symmetry or conic construction, singles out a classical configuration. This is an instance of a broader question: when does the projective deep-hole locus determine an MDS code, and when can that locus be algebraically degenerate?

The main theorem starts from no symmetry or conic construction. Among all six-arcs of  $\text{PG}(2, 11)$ , conic containment of the projective deep-hole syndrome locus reconstructs the Clebsch hexagon and its  $A_5$  stabilizer. A line bound, a chord-defect identity, and Dye’s Brianchon theorem give the conceptual implication; the census is needed only for the numerical clause. We then prove quantitative gaps, reconstruct the Brianchon geometry from decoder ambiguity, and isolate  $q = 11$ . For a Clebsch hexagon  $K$ , the same chord identity gives  $|\mathcal{U}(K)| = q^2 - 14q + 45$ , with off-conic excess  $(q - 4)(q - 11)$  when  $q \equiv 3 \pmod{4}$ . In general the same moment calculation shows that a conic-filling  $k$ -arc must satisfy  $q < \binom{k}{2}$ ; we classify the cases through seven points and reduce the eight-point case to four fields.

The configuration goes back to Clebsch [3, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [4, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [12, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [18]. Deep holes of Reed–Solomon codes have both an algorithmic history [13, 6] and a geometric classification in the projective case [19]. Those results concern GRS codes, whose arcs lie on normal rational curves; the code here is not GRS.

**What is new and what is not.** The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [3, 4, 12, 1, 18]. Sadeh classified the six-arcs of  $\text{PG}(2, 11)$  [16]; we claim no priority for that census or its extension counts. We prove the conic-locus identification of Proposition 3.2 directly here. The new contribution is the symmetry-free coding criterion and its consequences: low-degree rigidity, quantitative gaps, leader-support bipartition, decoder reconstruction, and the finite-field boundary.

## 2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C} : XZ = Y^2$$

in  $\text{PG}(2, 11)$ , with its 12 points identified with  $\text{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$  in the usual way. The stabilizer of  $\mathcal{C}$  in  $\text{PGL}(3, 11)$  is  $\text{PGL}_2(11)$ , acting on  $\mathcal{C}$  as it acts on the projective line.

A convenient representative is

$$A = \{(1, 10, 0), (1, 9, 1), (1, 4, 7), (1, 8, 5), (0, 1, 4), (1, 1, 7)\}.$$

It is a six-arc disjoint from  $\mathcal{C}$ ; its fifteen joins also avoid  $\mathcal{C}$ . The following invariant construction explains its symmetry.

**Definition 2.1** (Clebsch hexagon). Let  $A_5 \subset \mathrm{PGL}_2(11)$  be an icosahedral subgroup. The six Sylow 5-subgroups of  $A_5$  each fix two points of  $\mathcal{C}$ . The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an external, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are external, and those joins concur three at a time at ten internal Brianchon points [4, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [1, pp. 143,146]. Their  $q = 11$  list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only  $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$ ; the arc hypothesis selects the Clebsch branch, and Proposition 3.2 proves equality.

This is the arc  $K_2$  of Storme and Van Maldeghem [18, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [12, Theorem 1]. Dye proves uniqueness, stabilizer  $A_5$ , and the unique associated polarity; in the displayed coordinates its conic is  $\mathcal{C}$ . His calculation also shows that the six vertices lie on no conic in characteristic 11 [12, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at  $q = 11$  [18, Prop. 13], so the associated redundancy-three code has covering radius three.

### 3 The code and its projective syndrome locus

Take the parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 10 & 9 & 4 & 8 & 1 & 1 \\ 0 & 1 & 7 & 5 & 4 & 7 \end{pmatrix},$$

whose columns are the six displayed points of  $A$ , and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^\top = 0\}$$

be the resulting  $[6, 3, 4]_{11}$  MDS code. Since the six columns of  $H$  do not lie on any conic (their evaluation matrix against the six quadratic monomials  $X^2, XY, XZ, Y^2, YZ, Z^2$  is nonsingular),  $C$  is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [17].

The arc-coset dictionary from the introduction applies without a GRS hypothesis [7, Def. 6.1, Thm. 6.3]. Each uncovered direction  $x$  lifts to  $q - 1$  weight-three cosets, and each has  $\binom{6}{3} = 20$  minimum-weight leaders. Geometrically, the same condition says that adjoining  $x$  as a seventh column preserves the MDS property.

For example,  $s = (1, 0, 0)$  lies on  $\mathcal{C}$  and is therefore uncovered. The error

$$e = (7, 4, 1, 0, 0, 0)$$

satisfies  $He^\top = s$ . No error of weight one or two has syndrome  $s$ , while the twenty triples of coordinate positions each support one weight-three leader. Thus this single syndrome exhibits the complete dictionary: its projective direction is a conic point, its coset has distance three, and its nearest-codeword ambiguity has size twenty.

The stabilizer already forces the small point sets that will occur.

**Proposition 3.1** (The  $A_5$  point orbits). *Let  $G = \text{Stab}(A) \cong A_5$ . Its orbits on  $\text{PG}(2, 11)$  have lengths*

$$6, 10, 12, 15, 30, 30, 30.$$

*The first four are respectively  $A$ , the ten Brianchon points,  $\mathcal{C}(\mathbb{F}_{11})$ , and the fifteen vertices of the five self-polar triangles. In particular,  $\mathcal{C}(\mathbb{F}_{11})$  is the unique twelve-point  $G$ -orbit.*

*Proof.* Dye's polarity and stabilizer theorems identify  $G$  with the ternary icosahedral representation preserving  $\mathcal{C}$ , and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [12, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit–stabilizer to recover the orbit lengths.

Since  $11 \nmid 60$ , the representation is semisimple, and the ternary icosahedral representation is irreducible; hence  $G$  fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in  $\mathbb{F}_{11}$ , shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in  $\mathbb{F}_{11}$ ; and the three order-five eigenvalues are distinct because  $5 \mid 10$ . Restricting the same representation to the standard subgroups of  $A_5$  gives the following common-fixed-point ledger; the standard subgroup classification of  $A_5$  shows that these are all possible nontrivial proper point stabilizers. Here  $D_5$  has order ten, and the last row counts points whose stabilizer is exactly the displayed subgroup type. For the noncyclic entries, determine the common projective eigenlines of standard generators: a  $D_5$  normalizer retains one of the three  $C_5$  eigenlines, an  $S_3$  normalizer retains the unique  $C_3$  eigenline, a  $V_4$  has three simultaneous eigenlines, and its overgroup  $A_4$  permutes those three and fixes none. This gives the second row.

| $H$                                     | $A_4$ | $D_5$ | $S_3$ | $C_5$ | $V_4$ | $C_3$ |
|-----------------------------------------|-------|-------|-------|-------|-------|-------|
| number of conjugate subgroups           | 5     | 6     | 10    | 6     | 5     | 10    |
| fixed points for one $H$                | 0     | 1     | 1     | 3     | 3     | 1     |
| points with stabilizer conjugate to $H$ | 0     | 6     | 10    | 12    | 15    | 0     |

By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15; only points with exact stabilizer  $C_2$  remain. To spell out the subtraction in the last row: the normalizer  $D_5$  fixes one of the three  $C_5$  eigenlines and interchanges the other two, giving 6 points with stabilizer  $D_5$  and 12 with stabilizer  $C_5$ ; the unique  $C_3$ -fixed line is already fixed by its  $S_3$  normalizer; and  $A_4$  cycles the three  $V_4$  eigenlines, leaving all fifteen with exact stabilizer  $V_4$ .

It remains to account for exact  $C_2$  stabilizers. Each involution lies in two  $D_5$  subgroups, two  $S_3$  subgroups, and one  $V_4$ . Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer  $C_2$ . There are fifteen involutions, so this gives 90 points, necessarily three orbits of length  $60/2 = 30$ . The total  $6 + 10 + 12 + 15 + 90 = 133$  exhausts the plane. Dye's Theorem 6(v), specialized at  $q = 11 \equiv 1 \pmod{10}$ , identifies the twelve-point  $C_5$ -stabilizer orbit with  $\mathcal{C}(\mathbb{F}_{11})$  [12, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above.  $\square$

**Proposition 3.2** (The uncovered locus is the conic). *The covering radius of  $\mathcal{C}$  is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(\mathcal{C}) = \mathcal{C}(\mathbb{F}_{11}).$$

*Proof.* The set  $\mathcal{U}(A)$  is  $G$ -invariant. Dye's Clebsch classification gives exactly ten Brianchon points. The relevant double count is short. The fifteen edges contribute 150 off-arc incidences. The 45 pairs of disjoint edges meet once, so if  $n_i$  counts off-arc points lying on exactly  $i$  edges, then a point on three edges accounts for  $\binom{3}{2} = 3$  such pairs and

$$n_1 + 2n_2 + 3n_3 = 150, \quad n_2 + 3n_3 = 45, \quad n_3 = 10.$$

Thus the edges cover  $n_1 + n_2 + n_3 = 115$  of the 127 off-arc points, leaving  $|\mathcal{U}(A)| = 12$ . This is the  $q = 11$  specialization of the chord-defect identity in Lemma 6.1. By definition  $\mathcal{U}(A)$  is disjoint from the unique six-point orbit  $A$ . The orbit lengths in Proposition 3.1 therefore force this invariant twelve-set to be the unique twelve-point orbit  $\mathcal{C}(\mathbb{F}_{11})$ . By the dictionary above this is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. Dye's edge criterion and the complete-exterior-set description of Blokhuis–Seress–Wilbrink independently supply the classical inclusion  $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$  [12, discussion preceding Theorem 6, p. 281]; see also [1, pp. 143, 146].  $\square$

**Corollary 3.3** (Directions, cosets, leaders, and received words).  *$\mathcal{U}(A)$  has 12 directions, exactly the points of  $\mathcal{C}$ . Above them lie  $12(11 - 1) = 120$  nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has  $\binom{6}{3} = 20$  minimum-weight leaders, for 2400 leaders in all, and contains  $|C| = 11^3 = 1331$  received words. Consequently  $C$  has  $120 \cdot 1331 = 159720$  received-word deep holes.*

*Proof.* Immediate from Proposition 3.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of  $\mathbb{F}_{11}^3 \setminus \{0\} \rightarrow \text{PG}(2, 11)$  over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has  $|C|$  words.  $\square$

### 3.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is  $A_5$  of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^\times \longrightarrow \text{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars. The sequence splits. Since  $\gcd(3, 10) = 1$ , each projective class has a unique determinant-one representative; their monomial lifts give an  $A_5$  complement. Hence

$$\text{MAut}(C) \cong C_{10} \times A_5$$

has order 600. For the displayed column representatives, the subgroup of pure coordinate permutations is trivial.

**Proposition 3.4** (One orbit on received-word deep holes). *The monomial automorphism group of  $C$  is transitive on the 120 deep-hole cosets. Let  $\Gamma$  be the group generated by these automorphisms and by translations  $v \mapsto v + c$  for  $c \in C$ . Then*

$$|\Gamma| = |C| |\text{MAut}(C)| = 1331 \cdot 600 = 798600,$$

*and  $\Gamma$  is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.*

*Proof.* The projective stabilizer  $A_5$  is transitive on the twelve points of  $\mathcal{C}$ . Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust  $\mathcal{C}(\mathbb{F}_{11})$ . Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above  $\mathcal{C}$ . Its central scalar subgroup  $\mathbb{F}_{11}^\times$  acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all  $12 \cdot 10 = 120$  maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes  $v$  and  $w$ , choose  $m \in \text{MAut}(C)$  with  $\sigma(mv) = \sigma(w)$ . Then  $c = w - mv$  has zero syndrome, so  $c \in C$ , and the codeword translation by  $c$  sends  $mv$  to  $w$ . Both kinds of generator are Hamming isometries preserving  $C$ . The monomial group normalizes the translation subgroup, since  $m t_c m^{-1} = t_{m(c)}$  with  $m(c) \in C$ . Thus  $\Gamma = C \rtimes \text{MAut}(C)$ ; the two subgroups have trivial intersection and the displayed product order follows. Orbit-stabilizer now gives stabilizer order  $798600/159720 = 5$ , and every group of prime order is cyclic.  $\square$

### 3.2 Decoding consequences

The coset-leader-weight distribution of  $C$  is  $(1, 60, 1150, 120)$  (cosets of weight 0, 1, 2, 3, summing to  $11^3 = 1331$ ); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is  $(1, 0, 0, 0, 150, 420, 760)$ , forced by the MDS parameters. This is the radius-three instance of the coset-weight framework of Davydov, Marcugini, and Pambianco [7, §6]; the secant indices below refine its weight-two stratum by nearest-word multiplicity.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as  $s = (s_0, s_1, s_2)$  and put  $Q(s) = s_0 s_2 - s_1^2$ . For a point  $x \notin A$ , its *secant index* is the number of chords of  $A$  through  $x$ ; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray  $x$ .

**Proposition 3.5** (Complete syndrome oracle and ambiguity enumerator). *For  $v \in \mathbb{F}_{11}^6$  with syndrome  $s = Hv^\top$ ,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

*Among the 1330 nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are*

$$960, \quad 150, \quad 100, \quad 120.$$

*For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.*

*Proof.* The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 3.2 identifies the last set with  $Q = 0$ , proving the oracle. In the notation of its proof, the chord-intersection equations give  $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$ , where  $n_i$  counts projective directions of secant index  $i$  and  $n_0$  is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add

sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders.  $\square$

### 3.3 Self-polar structure

Edge’s finite-plane construction [4, §§30.2–31] and Dye’s general-field synthetic theory [12, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for  $\mathcal{C}$ . Their vertex-pair matchings partition the fifteen pairs into a *synthematic total*, that is, a partition of the edges of  $K_6$  into five perfect matchings. Section 5 turns its ten pairs of matchings into the Brianchon–support dictionary.

## 4 The rigidity theorem

Proposition 3.2 concerns one known arc. The following theorem is the paper’s main claim: among *all* six-arcs of  $\text{PG}(2, 11)$ , the property “the uncovered locus lies on a conic” picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

**Two conditions, one word apart.** The condition studied here is that  $\mathcal{U}(A)$  — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc  $A$  itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are mutually exclusive for six-arcs in  $\text{PG}(2, 11)$ : Lemma 4.1 gives the exact four-value subcensus for six-arcs lying on a conic, so none belongs to the Clebsch class in clause (iii) of Theorem 4.2.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

The external-line case is a six-point instance of the nonexistence in odd order of non-trivial *hyperfocused arcs*, whose secants determine only  $k - 1$  points on an external line [2, 5, 15]. We give a direct proof because the three possible line positions are then handled together.

**Lemma 4.1** (Line bound and conic-inscribed subcensus). *Let  $A$  be a six-arc in  $\text{PG}(2, q)$ , where  $q$  is odd. Every line contains at most  $q - 5$  points of  $\mathcal{U}(A)$ . Moreover, if  $q = 11$  and  $A$  lies on a nonsingular conic, then*

$$|\mathcal{U}(A)| \in \{18, 19, 20, 22\}.$$

*Proof.* A chord contains no point of  $\mathcal{U}(A)$ . If a non-chord line  $\ell$  contains one vertex, the ten chords on the other five vertices meet  $\ell$  in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off  $\ell$ ; their intersections with  $\ell$  therefore give a proper edge-colouring of  $K_5$ , whose chromatic index is five. Together with the vertex, at least six of the  $q + 1$  points of  $\ell$  are therefore outside  $\mathcal{U}(A)$ .

It remains to consider  $\ell \cap A = \emptyset$ . The fifteen chords meet  $\ell$  in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on  $\ell$ , so all chords through one point of  $\ell$  have disjoint endpoint pairs.

Hence there are at least five covered points. Equality would give a proper five-edge-colouring of  $K_6$ , necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \quad 02|14|35, \quad 03|15|24.$$

Send  $\ell$  to infinity and normalize the first two directions to horizontal and vertical. Put  $p_0 = (0, 0)$ ,  $p_1 = (x, 0)$  and  $p_2 = (0, y)$ , where  $xy \neq 0$ . Writing  $p_3 = (a, y)$ , the first two factors force  $p_4 = (x, b)$  and  $p_5 = (a, b)$ . Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is  $2xy = 0$ , impossible for odd  $q$ . Thus a disjoint line has at least six covered points, and in all cases at most  $(q + 1) - 6 = q - 5$  points of  $\mathcal{U}(A)$  remain.

The final assertion is the conic-inscribed subcensus of the independent rigidity replay listed in Table 3.  $\square$

**Theorem 4.2** (Rigidity theorem). *For a six-arc  $A \subset \text{PG}(2, 11)$ , the following are equivalent:*

- (i) *the uncovered locus  $\mathcal{U}(A)$  lies on some conic (possibly degenerate);*
- (ii)  *$\mathcal{U}(A)$  is exactly the set of  $\mathbb{F}_{11}$ -points of a nonsingular conic;*
- (iii)  *$A$  is  $\text{PGL}(3, 11)$ -equivalent to the Clebsch hexagon;*

*Proof.* Suppose first that  $\mathcal{U}(A)$  lies on a conic  $Q$ . If  $Q$  is nonsingular then  $|\mathcal{U}(A)| \leq q + 1 = 12$ . If  $Q$  is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 4.1 again gives  $|\mathcal{U}(A)| \leq 12$ . On the other hand, the chord-defect identity of Lemma 6.1 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where  $c(A)$  is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon  $A$ . Over a field of characteristic different from two, Dye proves that every six-arc has at most ten Brianchon points and that the equality cases form a single  $\text{PGL}_3(K)$ -orbit [12, §2.2–2.3, Theorem 1, pp. 275–276]. This bound has no symmetry, conic, or genericity hypothesis beyond the six-arc assumption. Hence  $c(A) \leq 10$ , so  $|\mathcal{U}(A)| \geq 12$ ; equality holds throughout,  $c(A) = 10$ , and  $A$  is a Clebsch hexagon.

The degenerate-conic branch is therefore vacuous. Indeed, Proposition 3.2 identifies  $\mathcal{U}(A)$  with the twelve rational points of a nonsingular conic. A degenerate conic cannot contain those same twelve points: it has no component in common with the nonsingular conic, so Bézout's theorem bounds their intersection by four. Thus the original containing conic is nonsingular. This proves (i) $\Rightarrow$ (iii). Proposition 3.2 and projective equivariance give (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i). Dye's stabilizer theorem also identifies the stabilizer in these equivalent cases with  $A_5$ .  $\square$

Table 1 prints the complete census at the level used by the rigidity and low-degree arguments. Every representative contains the standard frame

$$(0, 0, 1), (0, 1, 0), (1, 0, 0), (1, 1, 1);$$

the table gives its remaining two points. Here  $\delta$  is the nearest-conic discrepancy and  $d_{\min}$  is the least degree of a nonzero form vanishing on the uncovered locus.



In the second column,  $xy/uv$  abbreviates the remaining points  $(1, x, y), (1, u, v)$ .

| class | pair  | $ G $ | $ \mathcal{U} $ | $\delta$ | $d_{\min}$ | class | pair  | $ G $ | $ \mathcal{U} $ | $\delta$ | $d_{\min}$ |
|-------|-------|-------|-----------------|----------|------------|-------|-------|-------|-----------------|----------|------------|
| C01   | 23/32 | 2     | 20              | 12       | 5          | C06   | 23/42 | 2     | 20              | 16       | 5          |
| C02   | 23/34 | 6     | 18              | 14       | 4          | C07   | 23/48 | 1     | 21              | 17       | 5          |
| C03   | 23/36 | 3     | 19              | 13       | 5          | C08   | 23/49 | 4     | 20              | 16       | 5          |
| C04   | 23/37 | 3     | 19              | 17       | 5          | C09   | 23/56 | 4     | 20              | 16       | 5          |
| C05   | 23/39 | 4     | 20              | 16       | 5          | C10   | 23/62 | 6     | 19              | 15       | 5          |

| class | pair      | $ G $ | $ \mathcal{U} $ | $\delta$ | $d_{\min}$ |
|-------|-----------|-------|-----------------|----------|------------|
| C11   | 23/67     | 12    | 16              | 12       | 4          |
| C12   | 23/74     | 5     | 22              | 18       | 6          |
| C13   | 23/89     | 6     | 18              | 14       | 4          |
| C14   | 23/(9,10) | 12    | 18              | 14       | 4          |
| C15   | 34/45     | 60    | 12              | 0        | 2          |

Table 1: The fifteen projective classes of six-arcs in  $\text{PG}(2, 11)$ . Class **C15** is the Clebsch class. Both replay programs regenerate the same canonical keys; the global-gap replay supplies the middle three columns, and the low-degree replay supplies  $d_{\min}$ .

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

**Proposition 4.3** (Low-degree algebraic rigidity). *For a six-arc  $A \subset \text{PG}(2, 11)$ , the uncovered locus  $\mathcal{U}(A)$  is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if  $A$  is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation  $Q$ , and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

*Computer-assisted proof.* For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on  $\mathcal{U}(A)$ . The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by  $Q$ , and multiplication by linear forms gives the full three-dimensional cubic kernel.  $\square$

**Corollary 4.4** (Monomial characterization). *Up to monomial equivalence,  $C$  is the unique linear  $[6, 3, 4]_{11}$  code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular,  $A_5$  is recovered from this coding condition, not assumed as a hypothesis.*

*Proof.* The columns of a parity-check matrix of such a code form a six-arc  $A$ , and covering radius three identifies its projective deep-hole syndrome locus with  $\mathcal{U}(A)$ . Proposition 4.3 makes  $A$  projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and Dye’s stabilizer theorem gives the final  $A_5$  statement.  $\square$

*Remark 4.5* (Sharpness of the degree threshold). The cutoff cannot be raised from three to four: one non-Clebsch class has an 18-point uncovered locus equal to the rational zero set of a quartic. It is normalized class **C02** in the low-degree replay of Table 3, which prints

the exact equation and verifies the equality point by point. No smoothness, irreducibility, or genus assertion is used here.

**Theorem 4.6** (Numerical gap). *The Clebsch class has  $|\mathcal{U}(A)| = 12$ , while every non-Clebsch six-arc has  $|\mathcal{U}(A)| \geq 16$ . Equivalently,  $|\mathcal{U}(A)| \leq 15$  characterizes the Clebsch class, and equality  $|\mathcal{U}(A)| = 12$  is its exact value. If  $|\mathcal{U}(A)| \geq 16$ , then  $\mathcal{U}(A)$  lies on no conic.*

*Computer-assisted proof.* Any ordered four points of a six-arc form a projective frame, uniquely normalized to  $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$ . Enumerating the remaining unordered pair subject to the arc condition gives 1548 normalized sets. Each embedded arc has  $\binom{6}{4} 4! = 360$  ordered-frame choices. For each choice, apply the unique normalizing projectivity, sort the resulting six normalized points, and encode that list lexicographically. Taking the least of the 360 lists removes the frame choice; two arcs have the same least list exactly when a projectivity carries one to the other. This canonical key gives fifteen classes, and the number of distinct normalized presentations of a class with stabilizer  $G$  is  $360/|G|$ . Their extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\},$$

with multiplicities

$$(6, 30, 150, 300, 630, 360, 72)$$

across the 1548 normalized representatives. The six presentations attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Theorem 4.2 supplies the final conic-containment assertion.  $\square$

*Remark 4.7* (Why this rigidity is not in the extension-locus literature). The set  $\mathcal{U}(A)$  is the classical extension locus, studied for sufficiently large arcs through Segre’s tangent-envelope method; see Ball and Lavrauw [8, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime, while Sadeh’s census uses six-arcs for a different classification problem. Neither direction records the conic rigidity detected here.

## 5 The invariant support bipartition

Label the six coordinates by  $0, \dots, 5$ . The 20 three-element coordinate supports form two complementary  $A_5$ -orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support bipartition* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 1.

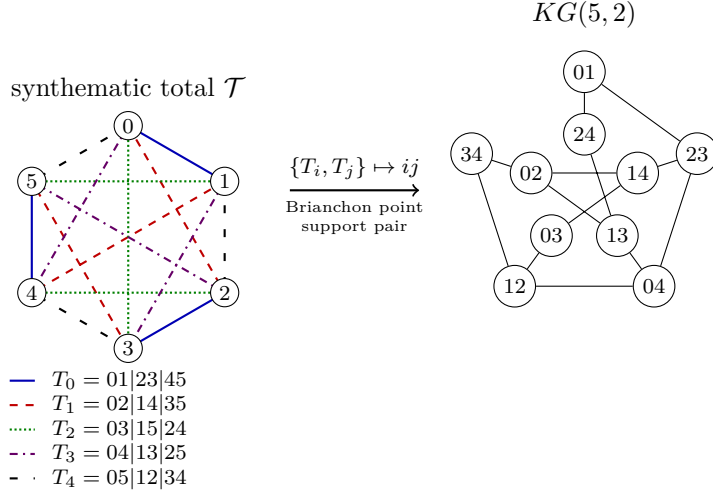


Figure 1: The synthematic–Petersen dictionary. The five line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair  $\{T_i, T_j\}$ , their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex  $ij$ ; adjacency means that the index pairs are disjoint.

**Proposition 5.1** (Brianchon–support dictionary). *Let  $\mathcal{T}$  be the five self-polar triangles of Edge and Dye [4, 12], regarded as the five perfect matchings in their synthematic total, and let  $\mathcal{B}(A)$  be the set of ten Brianchon points of the Clebsch hexagon. For distinct  $T, T' \in \mathcal{T}$ , the union  $T \cup T'$  is an alternating six-cycle. Let  $M(T, T')$  be the perfect matching joining opposite vertices of this cycle. Let  $\beta(T, T') = \{S, S^c\}$  be the unordered pair of alternating vertex classes of the cycle. For example,  $T_0 \cup T_1$  is the cycle  $0, 1, 4, 5, 3, 2$ , so  $M(T_0, T_1) = 05|13|24$  and  $\beta(T_0, T_1) = \{\{0, 3, 4\}, \{1, 2, 5\}\}$ .*

*The three chords indexed by  $M(T, T')$  concur at a point  $b(T, T') \in \mathcal{B}(A)$ . The maps*

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \quad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^c\} : |S| = 3\}, \quad \{T, T'\} \longmapsto \beta(T, T'),$$

*are  $A_5$ -equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of  $\mathcal{T}$ .*

*Proof.* The opposite-vertex matching is disjoint from  $T$  and  $T'$  and contains one edge from each of the other three members of the synthematic total. Hence  $\{T, T'\}$  is recovered as the two members of  $\mathcal{T}$  disjoint from  $M(T, T')$ , so the ten triangle pairs give exactly the ten perfect matchings outside the total. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge’s Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [4, §§19–20, 30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the invariant

total and incidence, both bijections and their compatibility with the support map are  $A_5$ -equivariant.  $\square$

**Corollary 5.2** (The decoder reconstructs the Brianchon configuration). *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five-matching synthematic total.*

*Proof.* The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 5.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings complementary to the total.  $\square$

**Proposition 5.3** (Invariant support bipartition). *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary  $A_5$ -orbits  $\mathcal{O}_+$  and  $\mathcal{O}_-$  of size ten. For every maximum-distance syndrome  $s$ , each support determines exactly one weight-three leader of the coset  $H^{-1}(s)$ ; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as  $1200 + 1200$ .*

*Every monomial automorphism of  $C$  preserves this unordered bipartition. The other coset in the exotic normalizer  $N_{S_6}(A_5) \cong S_5$  exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.*

*Proof.* The degree-six action of  $A_5$  on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique synthematic total, also preserved by its full  $S_5$  normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome  $s$  and a three-support  $S$ . The three columns indexed by  $S$  are independent because  $A$  is an arc, so they give a unique coefficient vector representing  $s$ . None of its three coefficients can vanish: otherwise  $[s]$  would lie on a secant of  $A$ , contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support  $S$ . Finally, every monomial automorphism induces a support permutation in  $A_5$ , by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve  $C$ , even after coordinate rescaling, because its support permutations lie outside that quotient  $A_5$ .  $\square$

## 6 Why $q = 11$

Call an arc  $A$  *conic-filling* if  $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$  for some nonsingular conic  $\mathcal{Q}$ . The conic-filling syndrome phenomenon of Sections 3–4 is specific to  $q = 11$ , even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

**Theorem 6.1** (Universal chord defect). *Let  $A$  be a  $k$ -arc in  $\text{PG}(2, q)$  with  $k \geq 4$ . Let  $n_i$  count the off-arc points incident with exactly  $i$  chords, put*

$$r = \lfloor k/2 \rfloor, \quad t(A) = \sum_{i=3}^r \binom{i-1}{2} n_i,$$

and write  $m = \binom{k}{2}$ . Then

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1\right) - t(A),$$

and

$$0 \leq t(A) \leq 3\binom{k}{4} \frac{r-2}{r}.$$

If  $\mathcal{U}(A)$  is the rational point set of a nonsingular conic, then

$$q < \binom{k}{2}.$$

For  $k = 6$ ,  $t(A)$  is the number  $c(A)$  of triple-chord concurrence points, so

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \quad 0 \leq c(A) \leq 15.$$

Consequently, if  $|\mathcal{U}(A)| = q + 1$ , then

$$c(A) = (q-6)(q-9).$$

*Proof.* At a point off  $A$ , concurrent chords have pairwise disjoint endpoint pairs, so at most  $r$  chords occur. Counting chord–point incidences and pairs of disjoint chords gives

$$\sum_i in_i = m(q-1), \quad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

Since  $\binom{i}{2} = (i-1) + \binom{i-1}{2}$ , the number of covered off-arc points is  $m(q-1) - 3\binom{k}{4} + t(A)$ . Subtracting it from the  $q^2 + q + 1 - k$  off-arc points proves the identity. Moreover,

$$\binom{i-1}{2} \leq \frac{r-2}{r} \binom{i}{2} \quad (3 \leq i \leq r),$$

which proves the bound on  $t(A)$ .

If  $|\mathcal{U}(A)| = q + 1$ , there are  $q^2 - k$  covered off-arc points. Hence  $q^2 - k \leq m(q-1)$ . Were  $q \geq m$ , this would give  $0 \leq q(q-m) \leq k - m < 0$ , a contradiction. For  $k = 6$ , one has  $r = 3$ ,  $t(A) = n_3 = c(A)$ , and the displayed specialization follows; setting its left side equal to  $q + 1$  gives  $c(A) = (q-6)(q-9)$ .  $\square$

**Lemma 6.2** ( $q = 9$  polarity graph). *For a nonsingular conic  $\mathcal{C} \subset \text{PG}(2, 9)$ , let  $\Sigma$  be the graph on the 36 internal points in which*

$$P \sim Q \iff Q \in P^\perp$$

*for the conic polarity. Then  $\Sigma$  is the Sylvester graph, with intersection array*

$$\{5, 4, 2; 1, 1, 4\}.$$

*For distinct internal points  $P, Q$ , the line  $PQ$  is passant to  $\mathcal{C}$  if and only if  $P$  and  $Q$  are at distance two in  $\Sigma$ , and the graph joining pairs at distance two has clique number*

$$\omega(\Sigma_2) = 5.$$

*Proof.* Counting internal points on polar lines and their intersections gives  $b_0 = 5$ ,  $b_1 = 4$ ,  $b_2 = 2$ ,  $c_1 = c_2 = 1$ , and  $c_3 = 4$ , hence the displayed intersection array. This is the Sylvester graph [10, §13.1.2]; uniqueness for this intersection array is proved in [11, Thm. 6].

For distinct internal points  $P, Q$ , put  $R = P^\perp \cap Q^\perp = (PQ)^\perp$ . If  $PQ$  is passant, then its pole  $R$  is internal and adjacent to both  $P$  and  $Q$ ; since the intersection array has  $a_1 = 0$ , these vertices are at distance two. Conversely, a common neighbor of  $P$  and  $Q$  must equal  $R$ , so distance two makes the pole of  $PQ$  internal and hence makes  $PQ$  passant. The final value is  $\text{eq}_2(\Sigma) = 5$  in [9, Table 1].  $\square$

**Theorem 6.3** (Uniqueness of  $q = 11$ ). *Let  $q$  be a prime power. If a six-arc  $A \subset \text{PG}(2, q)$  satisfies  $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$  for some nonsingular conic  $\mathcal{C}$ , then  $q = 11$ . At  $q = 11$  every such  $A$  is projectively equivalent to the Clebsch hexagon.*

*Proof.* Lemma 6.1 gives  $c(A) = (q - 6)(q - 9)$  with  $0 \leq c(A) \leq 15$ . The prime-power condition therefore leaves only  $q \in \{4, 5, 9, 11\}$ : at  $q = 2, 3$  the displayed value is, respectively, 28 and 18; the values at 7 and 8 are negative; 6 and 10 are not prime powers; and every integer  $q \geq 12$  gives  $c(A) \geq 18$ .

At  $q = 4$ , every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so  $\mathcal{U}(A)$  is empty. At  $q = 5$ , every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again  $\mathcal{U}(A)$  is empty.

It remains to exclude  $q = 9$ . Since every point of  $\mathcal{C}$  is uncovered, no chord of  $A$  meets  $\mathcal{C}$ ; every chord is therefore passant. A point off  $\mathcal{C}$  lies on five passants if it is internal and four if it is external, so the five chords through each vertex force all six vertices of  $A$  to be internal. Lemma 6.2 would then make  $A$  a six-clique in the distance-two graph of the Sylvester graph, whose clique number is five. This is impossible.

Therefore  $q = 11$ . The final assertion follows from Theorem 4.2, which puts every matching  $q = 11$  representative in the single Clebsch orbit.  $\square$

**Proposition 6.4** (The Clebsch-family uncovered formula). *Let  $K$  be a Clebsch hexagon over  $\mathbb{F}_q$ . Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

*If  $q \equiv 3 \pmod{4}$  and  $\mathcal{C}$  is Dye's associated conic, then  $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$  and*

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q - 4)(q - 11).$$

*In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if  $q = 11$ .*

*Proof.* Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [12, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by  $c(K)$ . Thus  $c(K) = 10$ , and Lemma 6.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of  $K$  are non-secants of the associated conic when  $q \equiv 3 \pmod{4}$  [12, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of  $\mathcal{C}$  is uncovered. Subtracting  $|\mathcal{C}(\mathbb{F}_q)| = q + 1$  from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at  $q = 4$  and  $q = 11$ , and the stated congruence leaves only  $q = 11$ . Conversely, Clebsch hexagons exist over  $\mathbb{F}_{11}$  by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points.  $\square$

*Remark 6.5* (The next admissible field). At  $q = 19$ , the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [12, p. 281], while Proposition 6.4 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus  $120 = (19 - 4)(19 - 11)$  additional points escape all fifteen secants, and  $\mathcal{U}(A)$  lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at  $q = 11$ .

### 6.1 The small-arc boundary

The chord moments also put the six-point theorem into a short classification for all smaller neighboring arc sizes.

**Theorem 6.6** (Conic-filling loci through eight points). *Let  $A$  be a  $k$ -arc in  $\text{PG}(2, q)$ , where  $q$  is a prime power and  $4 \leq k \leq 8$ , and suppose that  $\mathcal{U}(A)$  is the full set of  $\mathbb{F}_q$ -rational points of a nonsingular conic. If  $k \leq 7$ , then*

$$(k, q) = (4, 5) \quad \text{or} \quad (k, q) = (6, 11).$$

*In the first case  $A$  is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur. If  $k = 8$ , then*

$$q \in \{13, 16, 17, 19\}.$$

*Proof with exhaustive finite verification.* For  $k = 4$ , Lemma 6.1 gives  $|\mathcal{U}(A)| = (q-2)(q-3)$ . Equating this with  $q+1$  gives  $q = 5$ . Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where  $V(f)$  denotes the projective zero set of  $f$ ; this is a nonsingular six-point conic. For  $k = 5$ , the lemma gives  $|\mathcal{U}(A)| = q^2 - 9q + 21$ ; equality with  $q+1$  would require  $q^2 - 10q + 20 = 0$ , which has no integral root. The case  $k = 6$  is Theorem 6.3 together with Theorem 4.2.

For  $k = 7$ , Lemma 6.1 gives

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals  $q+1$ , then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The standard arc-size bound gives  $q \geq 7$ . The condition  $n_2 \geq 0$  gives  $q \leq 15$ , so the prime-power candidates are  $q \in \{7, 8, 9, 11, 13\}$ ; then  $n_1 \geq 0$  eliminates 7, 8, 9. Thus only  $q = 11$  and  $q = 13$  remain, with forced spectra  $(n_1, n_2, n_3) = (27, 78, 9)$  and  $(87, 60, 15)$  respectively. An exhaustive four-frame-normalized enumeration now supplies the finite exclusion. It finds 10232 distinct normalized seven-arcs at  $q = 11$ , of which 140 have  $|\mathcal{U}(A)| = q+1$ , and 53960 at  $q = 13$ , of which 1680 have that size. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted. For all 1820 candidates of the required size, the quadratic evaluation matrix has full rank six, so none lies on a conic.

Finally let  $k = 8$ . Here  $t(A) = n_3 + 3n_4$ , and Lemma 6.1 gives

$$t(A) = q^2 - 28q + 230, \quad t(A) \leq 105.$$

The second moment gives  $n_2 = 210 - 3t(A) + 3n_4$ . Since the number of covered off-arc points is  $q^2 - 8$ , the condition  $n_1 \geq 0$  then gives

$$n_4 \leq 28q - 448 + 3t(A).$$

The arc-size bound gives  $q \geq 7$ . The last inequality excludes  $q \in \{7, 8, 9, 11\}$ , while  $t(A) \leq 105$  excludes every  $q \geq 23$ . The remaining prime powers are exactly 13, 16, 17, 19.  $\square$

Thus conic filling is finite at every fixed arc size; the classification is complete through seven points, and the eight-point problem is reduced to four fields.

## 7 Verification architecture

Tables 2 and 3 separate the adopted claims, mathematical input, kernel-checked formal proof, and exhaustive finite enumeration. Here “kernel-checked” means that Lean’s trusted kernel checks the stated implication; executable replays instead enumerate the finite search space with exact arithmetic. The formal source tree supplied with the release artifact is pinned at commit `bf4fb39ab3c3b06c3f82c2c90d37077d7aa4c520`; it uses Lean 4.32.0-rc1 and Mathlib commit `571b8a8e54219b4d393f75f4b8653fac08197fcc`. Manifest, release, and executable checker paths are relative to this paper root; Lean paths are relative to the formal source root. These tables are the paper’s consolidated proof-mode ledger; the theorem statements and proofs do not carry separate “Proof mode” tags. The claim-by-claim manifest is `verification/trust_manifest.json`; the aggregate formal gate is `RelativeConicArcs/Gates/ClebschRigidityTrust.lean`. The clean replay entry point is `verification/verify_release.py`.

From the root of the archival Git-bundle clone, the exact replay commands are

```
cd papers/clebsch-rigidity
nix develop --command python3 verification/verify_release.py \
  --lean-root ../../lean
```

The SHA-256 digest of `verification/checker_outputs.json` is

`d9595a11734de303b21b3a214419e2c76668fa8b97353e44c914ef1077bc0b56`.

The digest of the aggregate Lean gate is

`dce51b42bfb384950bbffa756a87c3aabccfda03502d19545988f52aca32cb69`.



| Result                                          | Mathematical input                               | Kernel-checked formalization                                                                                                                                             |
|-------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_5$ orbits and syndrome conic                 | Dye; Proposition 3.1                             | <code>RelativeConicArcs/Q11A5PointOrbits.lean</code> ;<br><code>RelativeConicArcs/Q11Coding.lean</code>                                                                  |
| Decoding and support bipartition                | arc-coset dictionary; Edge–Dye triangles         | <code>RelativeConicArcs/Q11DecodingSynthesis.lean</code>                                                                                                                 |
| Rigidity theorem                                | line bound; chord defect; Dye’s Theorems 1 and 3 | <code>RelativeConicArcs/Q11DyeAxioms.lean</code> ;<br><code>RelativeConicArcs/Q11DyeConsequences.lean</code> ;<br><code>RelativeConicArcs/SixArcDefectBridge.lean</code> |
| Gap and low-degree loci                         | finite linear algebra over $\mathbb{F}_{11}$     | —                                                                                                                                                                        |
| Uniqueness of $q = 11$                          | chord moments; cited Sylvester bound             | <code>RelativeConicArcs/ClebschChordDefect.lean</code> ;<br><code>RelativeConicArcs/Q9Sylvester.lean</code>                                                              |
| Clebsch-family formula                          | Dye’s ten Brianchon points and edge criterion    | —                                                                                                                                                                        |
| Classification through seven; eight-point sieve | universal chord moments and defect bound         | <code>RelativeConicArcs/SmallKChordMoments.lean</code> ;<br><code>RelativeConicArcs/SmallKGeometricBridge.lean</code>                                                    |

Table 2: Mathematical and formal dependencies of the principal results.

The gap/low-degree result and the Clebsch-family formula have no Lean coverage. Their executable routes in Table 3 are load-bearing checks, not independent supplements to a formal proof.

| Result                            | Exhaustive or independent executable check                                                                                 |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| $A_5$ orbits and syndrome conic   | <code>check_rigidity_degenerate_conic.py</code>                                                                            |
| Decoding and support bipartition  | <code>check_decoding.py</code> ; <code>check_chirality.py</code> ;<br><code>check_code_automorphisms.py</code>             |
| Numerical gap and complete census | 1548-representative frame-normalized enumeration                                                                           |
| Gap and low-degree loci           | <code>check_global_conic_gap.py</code> ; <code>check_perturbation_gap.py</code> ;<br><code>check_low_degree_loci.py</code> |
| Uniqueness of $q = 11$            | <code>check_small_q_uniqueness.py</code> ; independent $q = 9$ construction                                                |
| Clebsch-family formula            | <code>check_q19_nonexample.py</code>                                                                                       |
| Classification through seven      | <code>check_small_k_conic_filling.py</code> ; the eight-point sieve is human algebra                                       |

Table 3: Executable checks, separate from the formal dependencies in Table 2.

The formal development axiomatizes exactly two consequences of Dye’s Theorem 1: the ten-point Brianchon bound and the classification of equality. The line bound and chord-defect identity are proved independently. At  $q = 11$ , the finite census also checks the imported numerical conclusion:  $|\mathcal{U}(A)| = 12 = 22 - 10$  is equivalent to the maximal value  $c(A) = 10$ .

## 8 Conclusion

The Clebsch code shows that covering-radius data can recover geometry that was not built into the code: conic containment of the deep-hole directions forces the Clebsch hexagon

and hence its  $A_5$  symmetry. The proof separates the mechanism from the census. A line bound and a chord-defect identity reduce the geometric condition to Dye’s equality case. Enumeration proves the low-degree strengthening, the sharp numerical gaps, and the terminal  $k = 7$  exclusions.

The same chord moments explain both the isolation of  $q = 11$  and the classification for  $4 \leq k \leq 7$ . For  $k \geq 8$ , further concurrency moments are needed. A second problem is to connect fixed- $k$  rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

## References

- [1] A. Blokhuis, Á. Seress, and H. A. Wilbrink, Characterization of complete exterior sets of conics, *Combinatorica* **12**(2) (1992), 143–147. doi:10.1007/BF01204717.
- [2] A. Bichara and G. Korchmáros, Note on  $(q + 2)$ -sets in a Galois plane of order  $q$ , in *Combinatorial and Geometric Structures and Their Applications*, Annals of Discrete Mathematics **14**, North-Holland, 1982, 117–121. doi:10.1016/S0304-0208(08)73588-6.
- [3] A. Clebsch, Ueber die Anwendung der quadratischen Substitution auf die Gleichungen 5-ten Grades und die geometrische Theorie des ebenen Fünfecks, *Mathematische Annalen* **4** (1871), 284–345.
- [4] W. L. Edge, Conics and orthogonal projectivities in a finite plane, *Canadian Journal of Mathematics* **8** (1956), 362–382.
- [5] D. A. Drake and K. Keating, Ovals and hyperovals in Desarguesian nets, *Designs, Codes and Cryptography* **31** (2004), 195–212. <https://doi.org/10.1023/B:DESI.0000015883.89954.80>; arXiv:math/0202115.
- [6] Q. Cheng and E. Murray, On deciding deep holes of Reed–Solomon codes, in *Theory and Applications of Models of Computation (TAMC 2007)*, Lecture Notes in Computer Science **4484**, Springer, 2007, 296–305.
- [7] A. A. Davydov, S. Marcugini, and F. Pambianco, On the weight distribution of the cosets of MDS codes, *Advances in Mathematics of Communications* **17**(5) (2023), 1115–1138. arXiv:2101.12722v2. (Numbered results are cited by their arXiv v2 numbering.)
- [8] S. Ball and M. Lavrauw, Arcs in finite projective spaces, *EMS Surveys in Mathematical Sciences* **6** (2019), 33–45. arXiv:1908.10772.
- [9] A. Abiad, A. Jabal Ameli, and L. Reijnders, The clique number of the exact distance  $t$ -power graph: complexity and eigenvalue bounds, *Discrete Applied Mathematics* **363** (2025), 55–70. doi:10.1016/j.dam.2024.11.032.
- [10] A. E. Brouwer, A. M. Cohen, and A. Neumaier, *Distance-Regular Graphs*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 18, Springer-Verlag, Berlin, 1989.
- [11] A. Jurišić and J. Vidali, The Sylvester graph and Moore graphs, *European Journal of Combinatorics* **80** (2019), 184–193. doi:10.1016/j.ejc.2018.02.018.
- [12] R. H. Dye, Hexagons, conics,  $A_5$  and  $\text{PSL}_2(K)$ , *Journal of the London Mathematical Society* (2) **44** (1991), 270–286. doi:10.1112/jlms/s2-44.2.270.

- [13] V. Guruswami and A. Vardy, Maximum-likelihood decoding of Reed–Solomon codes is NP-hard, *IEEE Transactions on Information Theory* **51**(7) (2005), 2249–2256.
- [14] J. W. P. Hirschfeld, *Projective Geometries over Finite Fields*, second edition, Oxford University Press, 1998.
- [15] S.-L. Ng and P. R. Wild, On  $k$ -arcs covering a line in finite projective planes, *Ars Combinatoria* **58** (2001), 289–300.
- [16] A. R. Sadeh, *Classification of  $k$ -arcs in  $\text{PG}(2, 11)$  and cubic surfaces in  $\text{PG}(3, 11)$* , D.Phil. thesis, University of Sussex, 1984.
- [17] L. Storme and J. A. Thas, Generalized Reed–Solomon codes and normal rational curves: an improvement of results by Seroussi and Roth, in *Advances in Finite Geometries and Designs*, Oxford University Press, 1991, 369–389.
- [18] L. Storme and H. Van Maldeghem, Primitive arcs in  $\text{PG}(2, q)$ , *Journal of Combinatorial Theory, Series A* **69** (1995), 200–216. doi:10.1016/0097-3165(95)90051-9.
- [19] Y. Zhang, D. Wan, and K. Kaipa, Deep holes of projective Reed–Solomon codes, *IEEE Transactions on Information Theory* **66**(4) (2020), 2392–2401. arXiv:1901.05445.